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# A Warehouse Floor at Scale

Three TurtleBots under Open-RMF Executing an  
Epistemic Policy over Forty Metres of Warehouse

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The companion report executes an epistemic mission on a floor of fourteen metres by twenty-one with two aisles. This one asks what changes when the floor is forty-two by sixty-three, holds a hundred and fifty-three obstacles and nine people walking through it, and is driven by three robots. Nothing in the epistemic layer changes. A great deal in the execution layer does, and most of this report is an account of what.

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## 1 Scope, and what this report is for

This is an integration result and not an epistemic one, and the distinction is stated here before anything else. The domain is unchanged: four policy nodes, AO\* to depth 3, forty expansions, two designated worlds, the same on the fourteen-by-twenty-one metre floor as here. Floor size and policy size are independent, and enlarging the warehouse from 296 to 2640 m<sup>2</sup> says nothing about the difficulty of the planning problem.

What is under examination is the execution layer between such a policy and a fleet, at a scale where the floor is no longer trivially traversable, and whether the properties the policy is chosen for — in particular, that one agent must *not* come to know something — remain checkable when it is not.

Two bounds are stated at the outset because they qualify everything after. The nine walking figures in the world are scenery: absent from the navigation graph, unperceived and unavoided. The private channel is topic addressing: the observer was not addressed on it, which is not the same as being unable to listen.

Two things follow. First, the substantive content is the lattice measurement of Section 2 and the failure analysis of Section 6, not the mission trace of Section 4: the mission is three actions long and its trace fits in a sentence. Second, a result of this kind is only as good as its stated bounds, and Section 7 sets out both halves: the four propositions the run establishes, and the four it does not.

## 2 The floor

### 2.1 Why an existing floor, and which

Open-RMF routes over a navigation graph and a building map, and its demonstrations ship both. The largest maps in `rmf_demos` are the airport terminal, at  $279 \times 60$  m with 92 named waypoints, and a campus at  $319 \times 246$  m; neither is a warehouse. There is no large warehouse map published for Open-RMF at all.

The candidate warehouse assets on Gazebo Fuel were therefore measured directly, from their collision meshes, not their catalogue descriptions. Table 1 records the result, and the discrepancy between what the descriptions promise and what the geometry provides is the substance of it.

Table 1: Warehouse assets on Gazebo Fuel, measured from their collision meshes.

| Asset                  | Footprint        | Collision         | Consequence                                                                                                                                    |
|------------------------|------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| OpenRobotics/Depot     | $29 \times 15$ m | a ground plane    | Racks and walls are visuals only. A laser passes through them and an occupancy grid rasterised from collision geometry returns an empty floor. |
| OpenRobotics/Warehouse | $30 \times 50$ m | shell and pillars | Genuinely large and genuinely empty: 1.2% occupied at robot height, all of it structural columns.                                              |
| industrial-warehouse   | as AWS small     | complete          | The AWS small warehouse recomposed from the same parts; no larger than the floor already in use.                                               |

Depot is the asset usually reached for — some fifty thousand downloads — and is the trap of the three. Its detail is entirely visual.

The floor finally used is `dynamic_logistics_warehouse`, which is AWS RoboMaker’s small warehouse tiled and furnished by hand: nine floor slabs, 153 obstacles drawn from fourteen AWS assets, five lamps, and nine scripted actors walking fixed circuits. Measured from its own model

poses it spans  $42.0 \times 62.8$  m, against  $14.05 \times 21.05$  m for the floor of the companion report — an increase of roughly nine times in area.

## 2.2 Deriving the roadmap

The companion report’s floor is parametric and its navigation graph is written from a formula. This one is not: nothing about the placement of 153 hand-set obstacles is regular, so the graph is derived from the world.

1. Every model in the world is one of fourteen AWS assets. Their collision extents are measured once, from the collision meshes, and cached.
2. Three of the fourteen are structure, not obstacle. Getting this wrong is not a detail: **GroundB** is a  $14 \times 21$  m floor tile, and treating it as an obstacle fills the warehouse solid.
3. The remaining models are rasterised at 0.25 m and inflated by the robot’s radius plus a margin, giving 0.45 m of clearance.
4. A lattice is laid over the free floor, lanes are kept only where the straight segment between their ends is clear, and the roadmap is then thinned.

The thinning is where the engineering is. A node is removed whenever every pair of its neighbours can be joined directly, which preserves every route and costs one waypoint; junctions and aisle waypoints are kept. Table 2 shows why both halves of that — fine lattice, then thinning — are necessary.

**Table 2:** Lattice spacing against connectivity and tractability.

| Spacing         | Free nodes | Largest component  | Outcome                        |
|-----------------|------------|--------------------|--------------------------------|
| 2.00 m          | 295        | 147 (50 %)         | floor splits into three bands  |
| 1.50 m          | 541        | 447 (83 %)         | still disconnected             |
| 1.25 m          | 748        | 748 (100 %)        | connected; planner intractable |
| 1.00 m          | 1205       | 1205 (100 %)       | connected; larger still        |
| 1.25 m, thinned | <b>261</b> | <b>261 (100 %)</b> | connected and routable         |

Six of the racks are eighteen metres long, so the floor is crossed by walls with few gaps. At two-metre spacing the lattice misses enough of those gaps to leave the warehouse in three disconnected north–south bands, and a largest-component filter then silently discards two thirds of the building. At 1.25 m every gap is found. The thinning recovers what the fine spacing costs: 261 waypoints, 35 of them named as aisles, and 1652 directed lanes.

## 2.3 Aisles are found, not listed

A waypoint is called an aisle when the floor runs out close on both sides along one axis and is open along the other. That is exactly the place a robot must enter before it can tell anything about what is inside, and so exactly what a sensing action wants to name. Thirty-five such waypoints survive a four-metre minimum separation, one per aisle, not one per cell.

## 3 The fleet

### 3.1 A TurtleBot3 that Open-RMF can drive

The fleet is three TurtleBot3 Waffles. Assembling one that RMF can actually drive took two corrections. Neither is documented anywhere obvious, so both are recorded here.

**The skeleton.** RMF moves a simulated robot with the `slotcar` plugin, which is a kinematic driver and not a controller. A Waffle built from scratch to what `slotcar` documents it requires — a `base_footprint`, a `left_wheel` and `right_wheel`, and the revolute joints `joint_tire_left` and

`joint_tire_right` — loaded without complaint, initialised slotcar, published its state at 6 Hz, accepted paths from the fleet adapter, and did not move a centimetre. On the same floor and the same graph, `rmf_demos`' TinyRobot drove sixty metres without trouble, which isolated the fault to the model, not the world or the roadmap. The Waffle geometry is therefore mounted on TinyRobot's drive skeleton unchanged.

**The scale.** `turtlebot3_gazebo` ships its meshes as STL in millimetres. Loaded at scale 1 a Waffle base is 271 metres across. The `.dae` files beside them in `turtlebot3_common` are placeholder cubes two metres on a side, which is a second and independent way of producing an enormous robot. Both were encountered.

### 3.2 Pinned allocation

Tasks are pinned to named robots through `robot_task_request` rather than dispatched for bidding. The reason is epistemic and not practical: the planner assigns actions to named agents while it solves, the model records knowledge against those names, and if the dispatcher were free to substitute a robot the model would credit an agent with having sensed something it never saw. An agent bound to no robot is a hard error and the bridge refuses the action.

## 4 The mission

The domain is the survey of `eplansys`, unchanged. Three agents; a site that may be contaminated; a goal of three conjuncts:

$$\gamma = Kw_{scout} \text{ contaminated} \wedge Kw_{relay} \text{ contaminated} \wedge \neg Kw_{observer} \text{ contaminated}.$$

The third conjunct is what gives the domain its difficulty. It forbids the open channel, which is otherwise the cheapest way for the team to inform itself, and so forces the planner to choose a channel by its audience. The planner returns a policy of four nodes, branching.

What the larger floor changes is the geography the policy is executed over. The site is `aisle_07`, some thirty metres from the robot's charger and reachable only across open floor: an aisle cannot be seen into until a robot stands at its mouth. That is the physical counterpart of the sensing action the policy contains, and it is the whole of what this floor supplies.

Timed from the opening of the recording, the run reads:

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|        |                                                             |
|--------|-------------------------------------------------------------|
| -64.0s | <code>goto-site_relay</code> dispatched; pre-positioning    |
| +9.5s  | <code>scan_relay</code> dispatched to <code>aisle_07</code> |
| +16.3s | observed <code>e-scan-dirty</code>                          |
| +18.5s | private announcement to <code>scout</code>                  |
| +21.4s | mission complete                                            |

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**The observation was injected, not sensed.** The log for this run reads `scan: RMF carried no outcome, falling back to the map's "e-scan-dirty"`. The value came from `default_outcome` in the task map. No sensor produced it and none is claimed. What follows that input ran as described: the product update was applied, the policy branched, the private channel carried the value, the goal was decided by model checking and the transcripts were read off the fleet. The injection is the one link in the chain that did not execute.

The claim is therefore narrow. The scan cost a thirty-metre transit and the entry of an aisle, where on the small floor it was discharged almost on dispatch. It is not a claim that a fleet worked 2640m<sup>2</sup>: one robot executed three actions across part of the floor while two others held station.

## 5 The epistemic layer

### 5.1 What is planned over

The domain is EPDDL and the logic is *S5* with three agents. The parser reports what it grounded before search begins:

```
[parser] Frame: S5 (knowledge)
[parser] Loaded: 4 atoms, 3 agents, 2 worlds (2 designated), 30 actions
          partial_obs=1 goal_kw_only=1
```

Two designated worlds states the uncertainty precisely. The initial model  $\mathcal{M}_0$  designates both a world in which the site is contaminated and one in which it is not, and no agent's accessibility relation separates them:

$$\mathcal{M}_0 \models \bigwedge_{i \in \mathcal{A}} \neg Kw_i \text{ contaminated}.$$

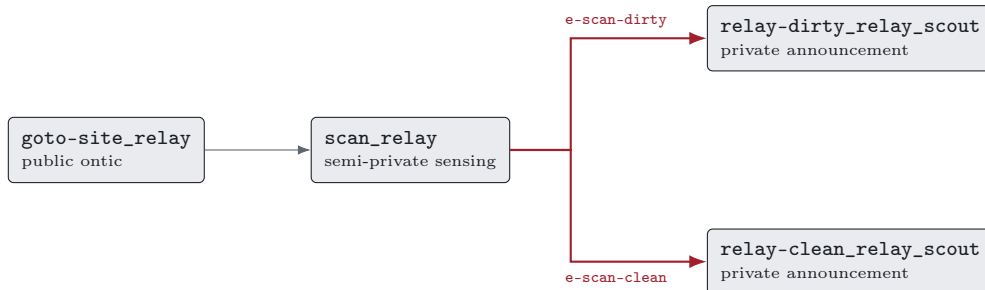
That the fleet does not know is a property of the structure, not a flag on a proposition, and it is the reason a set of facts will not serve as a representation.

The solver is plansys2/EpistemicPlanSolver, searching with AO\* because what it must return is not a sequence:

```
[epistemic] strategy=aostar heuristic=ks worlds=2 designated=2 actions=30
[aostar] Solution found at depth 3 Expanded=40 Generated=57 Memo=4/7
```

### 5.2 The policy, and why it branches

A classical planner returns a sequence because it presumes the state known. Here it is not, and the plan must prescribe an action for each outcome the sensing action admits. Figure 1 is what the solver returned.



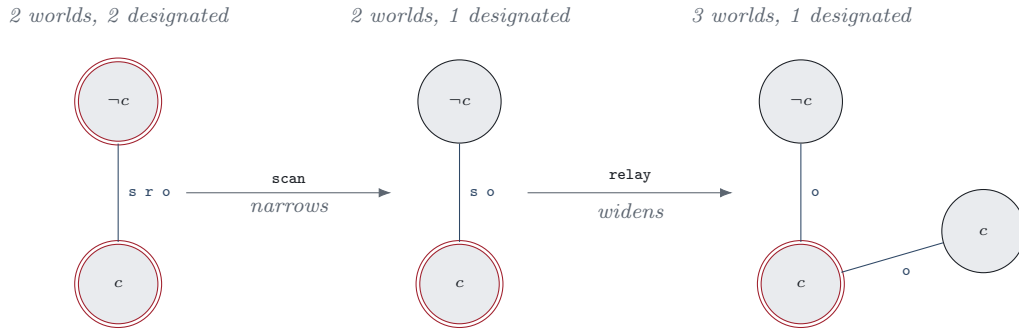
**Figure 1:** The four-node policy. The branch is on an observation, not on a state variable. Both arms announce privately.

That both arms are private is a consequence of the third conjunct and not a preference of the domain. A public announcement discharges  $Kw_{scout}$  and  $Kw_{relay}$  in one step and simultaneously falsifies  $\neg Kw_{observer}$ ; the planner is never instructed to prefer the private channel, only that the observer must not come to know.

### 5.3 The model under product update

Execution is not a matter of marking actions complete. Each executed action is an event model  $\mathcal{E}$ , and applying it forms the product  $\mathcal{M} \otimes \mathcal{E}$ . The state node reports the shape of the result at each step of the recorded run:

```
applied goto-site_relay: 2 worlds, 2 designated
applied scan_relay -> e-scan-dirty: 2 worlds, 1 designated
applied relay-dirty_relay_scout: 3 worlds, 1 designated
```



**Figure 2:** The model at each step. A double ring marks a designated world; an edge is labelled with the agents that cannot distinguish its endpoints. The counts are those the run reported.

The movement is in two directions and constitutes the argument for the representation.

**Sensing contracts the model.** Before the scan both worlds are designated; afterwards one is, the observation having excluded the other. The set of worlds is unchanged and the designation is not: what the agent learned is expressed by which worlds may still be actual.

**Private speech expands it.** Two worlds become three. An utterance the observer cannot hear introduces a distinction that did not previously exist — a world in which the message was sent, and a world the observer must go on considering possible in which it was not.

That these are the same operation is the point. A representation tracking only what is true could express neither: the first alters no proposition’s truth value, and the second increases the number of situations under consideration while the physical world stands still. The goal is decided against the resulting structure by model checking:

```
[goal] (and (Kw scout contaminated)
            (Kw relay contaminated)
            (not (Kw observer contaminated)))) holds
```

## 5.4 Where the chain is not closed

One qualification, and it is material. In the recorded run the outcome on which the policy branched was not sensed:

```
scan: RMF carried no outcome, falling back to the map's "e-scan-dirty".
      A fleet adapter reporting what it sensed would override this.
```

The value originated in the task map, not in a simulated sensor. Everything downstream of it — the product update, the branch, the private channel, the model checking — is genuine; its input was configured. The bridge emits that warning on every occurrence, which is correct for a stub that must not be mistaken for a measurement. Closing the link requires a fleet adapter that reports what it observed, or routing the sensing action through `plansys2_epistemic_perception`, which already translates occupancy geometry into propositions and which this pipeline presently bypasses.

## 6 Defects found by the integration

Putting the chain in operation at this scale revealed eight defects that no part of it exhibited alone. They are given in full. Taken together they are the substantive result of the exercise: six of the eight are consequences of composition, not errors of programming, and none would have appeared in a unit test.

**Table 3:** Defects encountered, in the order they were met.

| Symptom                                                                                   | Cause                                                                                                                                                                                         | Nature                 |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| No robot moves; the fleet manager is absent.                                              | The system <code>fastapi</code> is 0.63, written against pydantic 1; a pydantic 2 in <code>~/local</code> shadows the packaged 1.8 for every interpreter on the machine and the import fails. | Environment            |
| A robot leaves its charger, stops after 1.3 m, and RMF reports the task underway.         | A pallet jack placed for appearance overlapped the charger’s own lane by 20 cm. The graph check knew about racks and pillars and not about props.                                             | Composition            |
| The warehouse is a third of its size.                                                     | The nine floor slabs do not abut; a 12 cm seam reads as absent floor and splits the map into three components, of which a largest-component filter keeps one.                                 | Geometric              |
| No path is returned for a 60 m errand in 70 s.                                            | 693 waypoints and 2200 lanes. The fine lattice needed for connectivity is not the roadmap that should be routed on.                                                                           | Scale                  |
| Robots report a pose, accept paths and do not move.                                       | Spawned at $z = 0$ into floor slabs 12 cm thick with no ground plane beneath; each came to rest at $z = -0.067$ with its wheels inside the floor.                                             | Geometric              |
| A purpose-built TurtleBot3 loads, initialises slotcar, publishes state and does not move. | slotcar requires more of a model than it documents.                                                                                                                                           | Undocumented interface |
| The robot is 271 m across; or, separately, a 2 m cube.                                    | <code>turtlebot3_gazebo</code> meshes are STL in millimetres, and the <code>.dae</code> files beside them are placeholders.                                                                   | Units                  |
| <code>goto_site</code> fails with <i>RMF task timed out</i> .                             | The 180 s limit is the small floor’s figure. Forty metres at half a metre per second is not 180 s.                                                                                            | Scale                  |

Three of these deserve comment.

**The pallet jack.** The graph check was written to refuse lanes that cross obstacles, and it did refuse three layouts that looked correct on paper. It passed this one because its obstacle set was racks and pillars: the prop had been placed in the world generator, for appearance, and the checker never saw it. The symptom was severe out of all proportion to the cause — a robot pressed motionless into a pallet jack while RMF reported its task underway and nothing anywhere reported a collision. The correction moves the props into the definition the checker reads, and the check that missed it now rejects the original placement.

**The seam.** A twelve-centimetre gap between two floor tiles is invisible in any rendering and fatal to a rasteriser, which cannot distinguish a seam from a hole. The interaction with the largest-component filter is what makes it severe: the filter is correct in isolation — an isolated pocket of floor is a waypoint RMF will accept and never route to — and together they discard two thirds of a warehouse without a diagnostic.

**The lattice.** Connectivity and tractability pull in opposite directions, and the resolution that satisfies one fails the other. Building finely and thinning afterwards is not the same as building coarsely. Table 2 is the measurement.



## 7 Scope of the result

### 7.1 What the run establishes

Four propositions, in the existential register: there exists a run in which each held.

**The execution layer closes where traversal is not free.** A policy planned over an epistemic domain drove a fleet across 2640 m<sup>2</sup> of furnished warehouse, and the sensing action cost a thirty-metre transit and the entry of an aisle. It was not discharged on dispatch. On the small floor the scan was satisfied almost at the moment it was issued, the site being a waypoint the robot had effectively already reached. Here the two costs are of the same order.

**A negative epistemic conjunct was checked twice, independently.** Once against the model, by putting  $\neg Kw_{observer} contaminated$  to the epistemic state, and once against the fleet, by reading the radio transcript of the very agent the goal excludes. The second does not depend on the first being correct, and is the harder of the two to satisfy by accident.

**Pinned allocation held under RMF.** Every task was bound to the robot the planner named, so the agent the model credits with knowing is the agent whose laser returned the reading. No substitution occurred, and by construction none was available.

**A routable graph was recovered from a world not built to yield one.** 261 waypoints in a single connected component, derived from 153 hand-set obstacles by measurement alone, with no coordinate entered by hand.

### 7.2 What it does not establish

**The complexity of epistemic planning.** The policy has four nodes and would be solved as readily over a floor plan sketched on paper. Floor size and policy size are independent here, and enlarging the one says nothing about the other. A result of that kind requires a domain whose modal depth or agent count is varied deliberately.

**Behaviour under contention.** Three robots share the floor and three pinned tasks were accepted and executed concurrently without failure, but concurrency is not contention: nothing obliged two robots to occupy the same corridor at once. Settling the question means constructing that conflict on purpose and measuring what the schedule does with it — how long negotiation takes, whether it converges, and what it costs the mission that yields. The instrumentation exists; the experiment has not been run.

**Navigation among moving obstacles.** The world's nine actors walk fixed circuits and are absent from the navigation graph. The fleet neither perceives nor avoids them, and a robot meeting one would drive into it. They are scenery, and their presence in a recording should not be read as more than that.

**Confidentiality.** The private channel is realised by addressing: an utterance is published to a topic whose subscribers are the audience the action declares. Any node may subscribe to any topic. The claim available is that the team used a channel on which the excluded agent was not addressed, not that it could not have listened. The stronger claim requires DDS partitions or SROS 2 permissions, and those would be constructed from these same declared audiences, so what separates the two is enforcement, not design.

## 8 Conclusion

The execution layer holds at this scale, and the eight defects of Section 6 are the price of establishing it. Six are properties of the composition, not of any component. That is the recurring lesson of this line of work: a system of autonomous nodes, plugins loaded at run time and streams advancing at different rates exhibits a family of faults that integration alone can reveal.

The epistemic content is unchanged from the small floor, and deliberately so. What the larger floor supplies is a non-trivial geography. The site is thirty metres away and down an aisle, so the claim the domain exists to support, that an agent can be kept from coming to know something, is made against a floor on which movement costs something.